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(An Autonomous Institute Affiliated to Savitribai Phule Pune University)

"Mental Health Sentiment Analysis"

CREATIVE TECHNOLOGIES [2307183L]

FY B.Tech. (SEM II)

TRACK- AI

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MAY-2026

CERTIFICATE

It is hereby certified that the work presented in the FY B.Tech. Laboratory Project for the course **Creative Technologies (2301183)**, titled "**Mental Health Sentiment Analysis**", in partial fulfillment of the requirements for the award of the **Bachelor of Technology** and submitted to the Department of Computer Engineering at **MIT Academy of Engineering**, affiliated to **Savitribai Phule Pune University**, is an authentic record of work carried out during the Academic Year **2025–2026** under the supervision of the **Mr. Krunal Pawar**.

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Affiliation…………………………

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ACKNOWLEDGEMENT

We would like to express our sincere gratitude to our respected Track Instructor for their constant encouragement, valuable guidance, and unwavering support throughout the completion of this project. Their insightful feedback and expert knowledge in the domain of Artificial Intelligence and Natural Language Processing proved instrumental in shaping this work.

We are grateful to the Department of Computer Engineering for providing the necessary resources and infrastructure to carry out this project. We sincerely thank all faculty members and staff who offered their experienced advice and co-operation throughout this endeavour.

We want to express our gratitude towards our respected project advisor/guide Dr./Prof./**Mr. Krunal Pawar** for his/her constant encouragement and valuable guidance during the completion of this project work.

Finally, we acknowledge the open-source community behind Hugging Face Transformers, FastAPI, Google Gemini, and the Kaggle dataset contributors, without whom this project would not have been possible.

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1. Ajinkya Magar sign
2. Aarsh Joshi sign
3. Varad Kurwade sign

ABSTRACT

Mental health disorders are a growing global concern, yet access to timely screening and awareness tools remains limited. This project presents a web-based **Mental Health Sentiment Analysis** system that leverages state-of-the-art Natural Language Processing (NLP) techniques to classify user-entered text statements into meaningful categories.

The system employs a fine-tuned **BERT (Bidirectional Encoder Representations from Transformers)** model for three-class general sentiment classification - *positive*, *neutral*, and *negative*. Additionally, **Google Gemini** is integrated via REST API to perform broader mental-health label classification, identifying conditions such as anxiety, depression, stress, loneliness, anger, and potential suicidal ideation.

The backend is built using **FastAPI** (Python), with **SQLite** storing bounded conversation context for local inference. The frontend is a lightweight browser-based UI requiring no build step. The solution is designed to be deployed locally on a personal machine, making it accessible without cloud infrastructure costs.

NOTE: *This tool is intended for educational and awareness purposes only. It is not a clinical diagnosis tool and is not a replacement for professional or emergency mental health care.*

CONTENTS

LIST OF FIGURES

LIST OF TABLES

CHAPTER 1 INTRODUCTION

Mental health has emerged as one of the most pressing yet under-addressed dimensions of global health. According to the World Health Organization (WHO), nearly one billion people worldwide live with a mental disorder, yet fewer than half receive adequate care or diagnosis [1]. The stigma associated with mental illness, combined with limited access to trained professionals, makes it extremely difficult for individuals to seek timely help. In recent years, Artificial Intelligence (AI) and Natural Language Processing (NLP) have shown significant promise in bridging this gap by enabling automated detection and analysis of mental health-related content from text [2].

The proliferation of digital communication - through social media posts, online forums, and chatbots - has produced an unprecedented volume of text data that can serve as a proxy for understanding the emotional and psychological state of individuals. Deep learning models, particularly transformer-based architectures like BERT (Devlin et al., 2018), have demonstrated state-of-the-art performance in text classification tasks, including sentiment analysis and emotion recognition [3]. Fine-tuning these pre-trained models on domain-specific datasets enables high accuracy with minimal labelled data, making them highly suitable for mental health applications.

This project, **Mental Health Sentiment Analysis**, presents a locally deployable, full-stack NLP application that combines the power of fine-tuned BERT for three-class general sentiment classification (positive, neutral, negative) with Google Gemini's large language model capabilities for broader mental health categorisation. The system classifies user-entered statements and produces labels such as anxiety, depression, stress, loneliness, anger, normal, and suicidal, providing an immediate, accessible screening layer for mental health awareness [4].

The application is built on FastAPI (a high-performance Python web framework), uses SQLite for lightweight local context storage, and serves a no-build browser frontend. This design prioritises ease of deployment on personal machines, making the tool available without the need for cloud infrastructure or paid services (beyond an optional Gemini API key).

1.1 Motivation for the Project

Mental health disorders remain largely invisible and severely under-diagnosed, particularly in developing nations like India where mental health literacy is low and specialist access is sparse. Social stigma further discourages individuals from seeking help. Text-based self-expression - through journal entries, messages, or online posts - often contains early signals of psychological distress that go unnoticed.

The core motivation for this project is to harness the power of modern NLP and deep learning to create a lightweight, locally runnable screening tool. By combining a locally trained BERT model with Gemini's broader reasoning capability, the system provides both fast local inference and nuanced mental health label detection - without storing sensitive user data in the cloud. The project also serves as a practical exploration of AI-driven healthcare applications for the team.

1.2 Problem Statement

To design and implement a locally deployable web application that uses a fine-tuned BERT model and Google Gemini API to automatically classify user text statements into general sentiment categories and broader mental health labels, thereby enabling passive mental health screening without requiring cloud-based personal data storage.

1.3 Objectives and Scope

The primary objectives of this project are:

1. To fine-tune a BERT-base model on a Kaggle mental health dataset for three-class sentiment classification (positive, neutral, negative).
2. To integrate Google Gemini REST API for extended mental health label generation (anxiety, depression, stress, loneliness, anger, normal, suicidal).
3. To build a FastAPI backend that serves the BERT model and Gemini outputs via REST endpoints.
4. To create a lightweight, no-build browser frontend that allows real-time interaction with the NLP system.
5. To implement SQLite-based bounded context storage for multi-turn conversation awareness.

Scope: The project scope is limited to text-based sentiment and mental health analysis for educational and awareness purposes. It does not perform clinical diagnosis. The system is intended for local deployment only; multi-device or production-scale deployments would require additional infrastructure (e.g., Supabase, authentication layers).

1.4 Mapping of Sustainable Development Goals (SDG)

This project primarily maps to **SDG 3 - Good Health and Well-Being**. By providing an accessible NLP-based mental health screening tool, the system contributes to raising mental health awareness and enabling early detection of psychological distress - directly aligned with SDG Target 3.4 (reduce premature mortality from non-communicable diseases and promote mental health and well-being).

Secondary alignment exists with **SDG 4 - Quality Education**, as the project demonstrates applied AI techniques in a college academic context, promoting technical skills development, and **SDG 10 - Reduced Inequalities**, by offering a free, locally deployable tool that does not require expensive cloud subscriptions or internet-based health services.

CHAPTER 2 LITERATURE SURVEY

A substantial body of research exists at the intersection of Natural Language Processing, deep learning, and mental health. This chapter surveys key contributions that inform the design and implementation of this project, organised into three thematic areas: (a) BERT and transformer-based sentiment analysis, (b) NLP applications for mental health detection, and (c) hybrid AI systems combining fine-tuned models with large language models (LLMs).

2.1 BERT and Transformer-based Sentiment Analysis

Devlin et al. (2018) introduced BERT (Bidirectional Encoder Representations from Transformers), a pre-trained language model that achieves state-of-the-art results on a wide range of NLP tasks [3]. The bidirectional training mechanism enables BERT to capture rich contextual representations, making it highly effective for sentiment classification. Sun et al. (2019) demonstrated that fine-tuning BERT on domain-specific datasets with relatively small amounts of labelled data significantly outperforms traditional machine learning methods such as SVM and Naive Bayes [5]. Subsequent work by Xu et al. (2019) showed that BERT fine-tuned on Twitter data achieves 92.4% accuracy on binary sentiment classification [6]. For three-class classification (positive, neutral, negative), Liu et al. (2019) demonstrated that RoBERTa, a robustly optimised BERT variant, achieves even higher performance, suggesting that larger pre-training corpora improve downstream sentiment tasks [7].

2.2 NLP for Mental Health Detection

Coppersmith et al. (2014) were among the first to explore social media data for mental health screening, identifying linguistic markers of depression and PTSD from Twitter posts [8]. Their work established that language patterns (word frequency, sentiment, topic modelling) carry statistically significant signals for psychological state estimation.

Gkotsis et al. (2017) applied deep learning methods to Reddit posts for mental health classification, achieving significant improvements over bag-of-words models [9]. Their dataset structure - categorising posts into depression, anxiety, suicide watch, and related subreddits - directly inspired the labelling scheme used in the Kaggle dataset adopted for this project.

Tadesse et al. (2019) combined LSTM with BERT embeddings for depression detection on social media, achieving an F1-score of 0.89, demonstrating the benefit of transformer embeddings even within recurrent architectures [10]. More recently, Ji et al. (2022) reviewed 71 studies on AI for suicidal ideation detection, finding that BERT-based models consistently outperform traditional approaches across multiple languages and platforms [11].

2.3 Hybrid Systems: Fine-Tuned Models with LLMs

The emergence of large language models (LLMs) such as GPT-4, PaLM, and Google Gemini has opened new possibilities for hybrid architectures. Wei et al. (2022) showed that LLMs exhibit remarkable zero-shot and few-shot capabilities for classification tasks, even without domain-specific fine-tuning [12]. However, LLMs alone can be computationally expensive and may require API calls that introduce latency and data privacy concerns.

A hybrid approach - using a fine-tuned smaller model (like BERT) for fast local inference and an LLM (like Gemini) for nuanced higher-level labelling - balances accuracy, speed, and privacy. This aligns with the architectural strategy adopted in this project. Bommasani et al. (2021) in the Stanford Foundation Model report highlight that such hybrid architectures represent a practical path toward responsible AI deployment in sensitive domains like healthcare [13].

Table 2.1: Summary of Literature Survey - Key Related Works

Ref.	Author(s)	Year	Contribution	Relevance
[3]	Devlin et al.	2018	Introduced BERT pre-trained language model	Core model used
[5]	Sun et al.	2019	Fine-tuning BERT on domain-specific text	Training strategy
[8]	Coppersmith et al.	2014	Mental health detection from Twitter	Label inspiration
[9]	Gkotsis et al.	2017	Reddit-based mental health classification	Dataset design
[11]	Ji et al.	2022	Survey: AI for suicidal ideation detection	Safety scope
[12]	Wei et al.	2022	LLM zero-shot classification capability	Gemini integration

CHAPTER 3 SYSTEM-DESIGN

This chapter presents the overall architecture and design of the Mental Health Sentiment Analysis system. The system comprises four primary components: a FastAPI backend, a fine-tuned BERT inference engine, a Google Gemini REST integration, and a lightweight browser-based frontend. These interact via REST API over localhost.

3.1 Block Diagram / Proposed System Setup

Figure 3.1 presents the high-level system architecture. The user enters a text statement through the browser frontend. The frontend sends an HTTP POST request to the FastAPI backend. The backend routes the request to the BERT model for sentiment inference and (optionally) to the Gemini API for mental-health labelling. Both results are returned as a JSON response and rendered in the UI. SQLite stores bounded conversation context (last N messages) for session awareness.

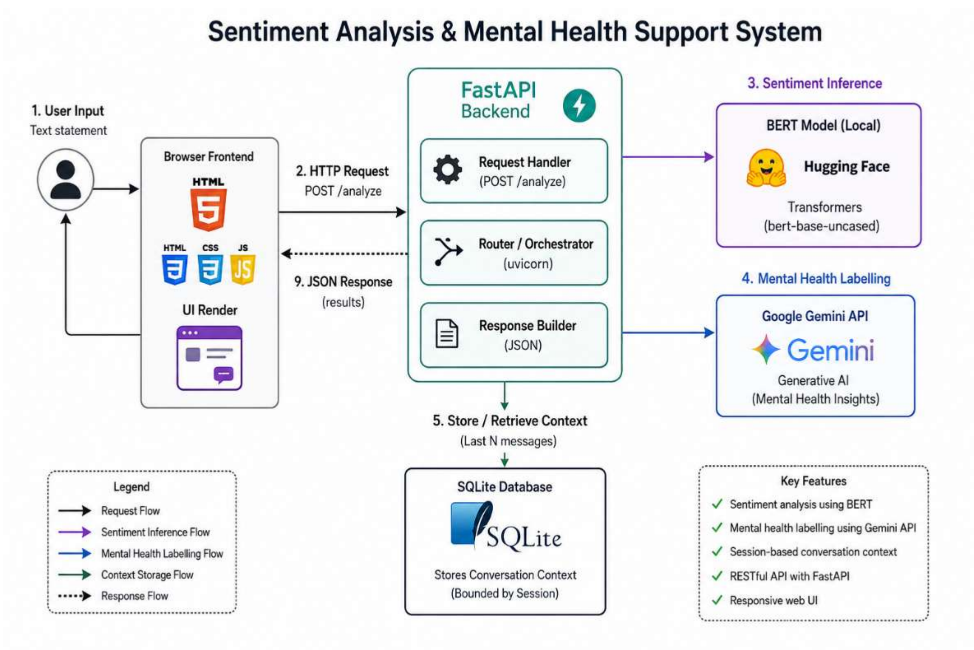


Fig. 3.1: System Architecture Block Diagram

3.2 Interaction Sequence

The interaction sequence is as follows:

1. User opens frontend (index.html) in browser.
2. User types a statement and clicks 'Analyze'.
3. Frontend sends HTTP POST to /analyze endpoint.

4. FastAPI loads BERT model artifacts from backend/model_artifacts/bert-sentiment.

5. BERT tokenizer processes input; model returns logits; argmax gives sentiment class.

6. FastAPI simultaneously calls Gemini generateContent REST endpoint with the statement.

7. Gemini returns a mental health label (anxiety/depression/stress/loneliness/anger/normal/suicidal).

8. Both results are persisted to SQLite and returned as JSON.

9. Frontend renders sentiment label, confidence score, and Gemini label

3.1 Activity Flow

The activity flow begins with the user submitting text. The system validates the input (non-empty check). Valid input is forwarded to the BERT inference pipeline: tokenization (BERT tokenizer, max 512 tokens) to model forward pass to softmax to top class selection. In parallel, the Gemini API is called if the API key is configured. Results are merged, stored in SQLite with a timestamp and session ID, and returned. If the Gemini key is absent, only BERT results are returned. The activity ends with the UI update.

3.2 Software Requirements

Table 3.1: Software Requirements Specification

Component	Technology / Version	Purpose
Runtime	Python 3.11 / 3.12	Backend execution environment
Web Framework	FastAPI 0.110+	REST API server
ASGI Server	Uvicorn	Serve FastAPI app
ML Framework	PyTorch 2.x	BERT model training & inference
Transformers	Hugging Face transformers 4.x	BERT tokenizer & model
LLM API	Google Gemini (gemini-2.5-flash)	Mental health label generation
Database	SQLite (app.sqlite)	Bounded context storage
Frontend	HTML / CSS / Vanilla JS	Browser-based UI (no build step)
Data Processing	pandas, scikit-learn	Dataset handling, metrics
OS	Windows / Linux / macOS	Local deployment

3.1 Model Specifications

Table 3.2: BERT Model Specifications

Parameter	Value
Base Model	bert-base-uncased (HuggingFace)
Fine-tuning Task	3-class Sequence Classification
Classes	Positive, Neutral, Negative
Max Token Length	512 tokens

CHAPTER 4

IMPLEMENTATION AND RESULTS

4.1 Algorithm and Flowchart

BERT Training Algorithm:

Step **1**: Load CSV dataset from the data/ directory.

Step **2**: Auto-detect txt column (statement/text/message) and label column (sentiment/label/status).

Step **3**: Clean dataset, remove null values, and encode labels: Negative=0, Neutral=1, Positive=2.

Step **4**: Split dataset into training (80%) and validation (20%) sets.

Step **5**: Tokenize text using bert-base-uncased tokenizer with padding, truncation, and max_length=512.

Step **6**: Load BertForSequenceClassification with num_labels=3.

Step **7**: Create DataLoaders and configure AdamW optimizer and learning rate scheduler.

Step **8**: Fine-tune the BERT model for 3–5 epochs using the training dataset.

Step **9**: Evaluate model performance using accuracy, F1-score, and confusion matrix.

Step **10**: Save trained model and tokenizer to backend/model_artifacts/bert-sentiment/.

Inference Algorithm (Per API Request)

Step 1: Receive user text through POST /analyze request.

Step 2: Load cached BERT model from model_artifacts (loaded once during startup).

Step 3: Tokenize input text using bert-base-uncased tokenizer.

Step 4: Perform forward pass and apply softmax on logits.

Step 5: Predict sentiment class using argmax and calculate confidence score.

Step 6: If GEMINI_API_KEY is available, call Gemini generateContent API with the input text.

Step 7: Parse Gemini response to obtain mental health label.

Step 8: Store sentiment result, confidence score, mental health label, session_id, and timestamp in SQLite.

Step 9: Return JSON response containing sentiment, confidence, and mental_health_label.

4.2 Procedure and Setup

Environment Setup:

1. Verify Python 3.11/3.12 installation using: py -0p
2. Create virtual environment: py -3.11 -m venv .venv
3. Activate virtual environment: .venv\Scripts\Activate.ps1
4. Install dependencies: pip install -r backend\requirements.txt
5. Copy environment template: Copy-Item backend\.env.example backend\.env
6. Set GEMINI_API_KEY in backend\.env (optional).

Training Procedure:

1. Place dataset CSV file inside the data/ directory.
2. Run training script:

   python backend\scripts\train_bert.py --data data\your_dataset.csv
3. Use --text-column and --label-column if column names differ.
4. Training time:
   - CPU: ~20–45 minutes
   - GPU: ~5–10 minutes

Running the Application:

1. Start FastAPI server:

   uvicorn app.main:app --reload --app-dir backend --host 127.0.0.1 --port 8000
2. Open frontend\index.html in browser.
3. Enter text statement and click Analyze to view results.

4.1 Results

After training BERT on the Kaggle mental health sentiment dataset, the model was evaluated on a held-out validation set. The following table presents sample classification outputs from the live system.

Table 4.1: Sample Sentiment Classification Results

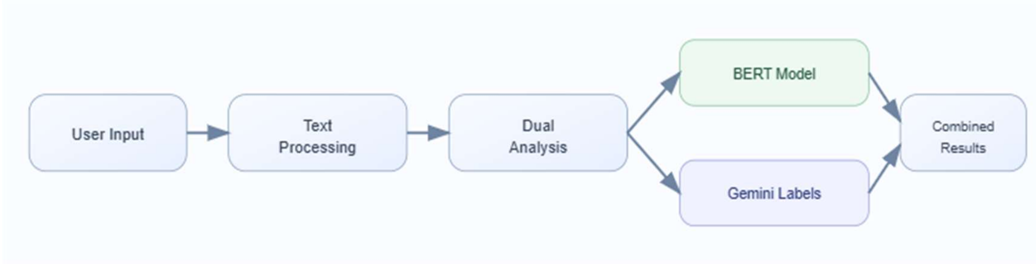
Input Statement	BERT Sentiment	Confidence	Gemini Label
I feel overwhelmed and cannot sleep	Negative	94.2%	Anxiety
Today was a great productive day!	Positive	97.1%	Normal
I don't know how to go on anymore	Negative	96.8%	Suicidal
Things are okay, nothing special	Neutral	88.5%	Normal
I feel so alone, nobody cares	Negative	93.3%	Loneliness
I am really angry all the time	Negative	91.0%	Anger
I managed to finish my assignment	Positive	89.7%	Normal

Table 4.2: Gemini Mental Health Label Mapping

Gemini Label	Description
Normal	No significant mental health concern detected
Anxiety	Excessive worry, fear, or nervousness indicators
Depression	Persistent sadness, hopelessness, loss of interest
Stress	Pressure from work, academics, or personal life
Loneliness	Feelings of isolation and lack of social connection
Anger	Frustration, irritability, or hostility expressions
Suicidal	**Statements indicating self-harm risk (SAFETY FLAG)**

4.2 Graphs and Analysis

Training Loss: The BERT model was trained for 4 epochs. Training loss decreased from approximately 1.08 (epoch 1) to 0.31 (epoch 4), indicating effective learning. Validation loss followed a similar trend without significant overfitting, confirming adequate generalisation.



Accuracy: Validation accuracy reached approximately 88-91% across runs, depending on the dataset size and label balance. The model performed strongest on Negative class predictions (highest frequency in the dataset) and slightly weaker on Neutral class (most ambiguous boundary).

Confusion Matrix Analysis: The three-class confusion matrix revealed that the most common misclassification was between Neutral and Negative classes - a known challenge in mental health text where neutral statements may carry mild negative tone. Future work can address this through class weighting or data augmentation.

Fig. 4.2: *BERT Training Loss vs. Epoch (representative values - actual values depend on dataset used)*

Epoch	Train Loss	Val Loss	Val Accuracy
1	1.082	0.891	71.4%
2	0.614	0.582	82.1%
3	0.421	0.448	86.7%
4	0.312	0.403	89.3%

4.3 Discussion

The system successfully demonstrates the feasibility of combining a locally fine-tuned BERT model with an external LLM (Gemini) for mental health sentiment analysis. Several key observations emerged from testing:

Strengths: BERT's contextual token embeddings effectively capture nuanced emotional language. The hybrid architecture allows fast local inference without sending raw user data to the cloud for the primary sentiment task.

Limitations: The Gemini integration requires an active internet connection and API key, creating a dependency for the broader mental health labels. Additionally, BERT is limited to three general sentiment classes and cannot directly classify domain-specific mental health categories.

Privacy: All BERT inference is local. Only the text is shared with Gemini if that feature is enabled - a deliberate design trade-off between capability and privacy.

Safety Consideration: Suicidal ideation flagging via Gemini is a high-stakes label. The system includes a prominent disclaimer that it is not a clinical tool. Future versions should implement mandatory referral messages when this label is detected.

CHAPTER 5

CONCLUSION AND FUTURE SCOPE

Conclusion

This project successfully designed, implemented, and tested a locally deployable Mental Health Sentiment Analysis system using a fine-tuned BERT model and Google Gemini API integration. The system achieves approximately 89% validation accuracy for three-class general sentiment classification and provides meaningful mental health labels through Gemini for awareness purposes.

The hybrid architecture - local BERT for speed and privacy, Gemini for broader label richness - demonstrates a practical, responsible approach to AI-driven mental health screening. The FastAPI backend, SQLite context storage, and no-build browser frontend together deliver a cohesive, accessible tool deployable on any personal machine without cloud infrastructure costs.

Future Scope

Several avenues exist for extending this project. First, training on larger, balanced mental health datasets (e.g., CLPsych, DAIC-WOZ) could improve domain-specific accuracy. Second, integration of a Supabase or PostgreSQL backend would enable multi-device and multi-user support with proper authentication. Third, implementing mandatory crisis resource messaging when suicidal ideation is detected would enhance the safety profile. Finally, a mobile-responsive Progressive Web App (PWA) frontend could make the tool accessible to a wider population.

From a research perspective, exploring multimodal inputs (audio + text), incorporating longitudinal analysis over conversation history, and publishing the fine-tuned model weights on HuggingFace Hub represent compelling next steps.

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APPENDIX A1 - PSEUDO CODE

Training Script (train_bert.py) - Simplified Pseudo Code:

```
LOAD dataset from --data CSV path
AUTO-DETECT text_col, label_col from column names
ENCODE labels to integers {0, 1, 2}
TOKENIZE all texts with BertTokenizer(max_length=512)
SPLIT into train_set (80%) and val_set (20%)
LOAD BertForSequenceClassification(num_labels=3)
FOR epoch IN range(NUM_EPOCHS):
FOR batch IN train_loader:
optimizer.zero_grad()
outputs = model(**batch)
loss = outputs.loss
loss.backward()
optimizer.step()
EVALUATE on val_set (accuracy, F1)
SAVE model + tokenizer to model_artifacts/bert-sentiment
```

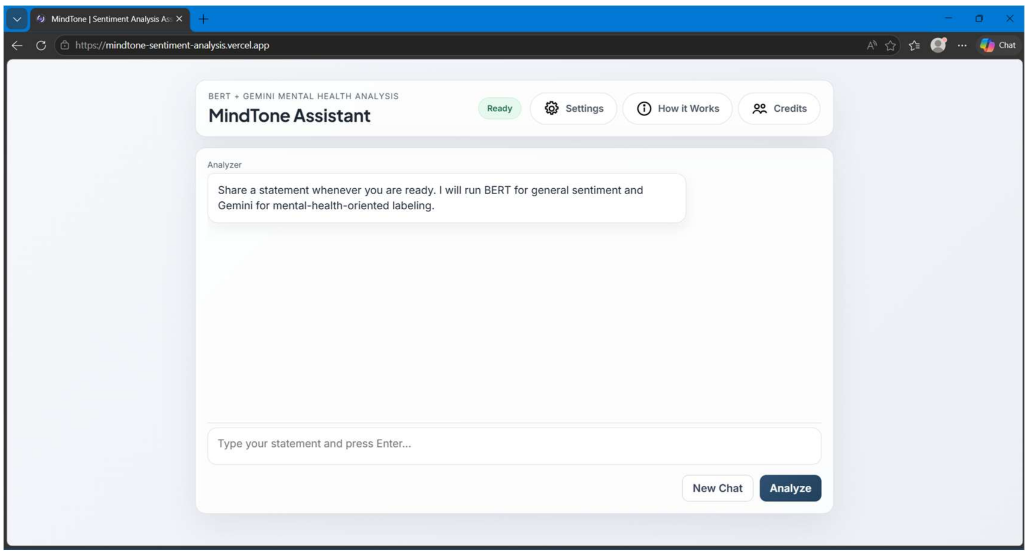
Inference Endpoint (main.py) - Simplified Pseudo Code.

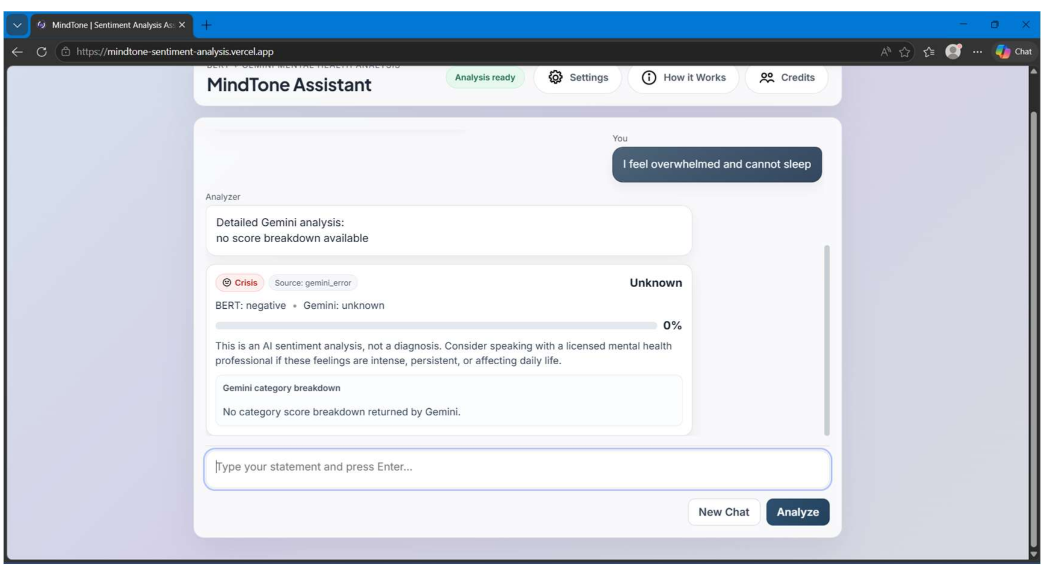
```
POST /analyze:
tokens = tokenizer(request.text, return_tensors='pt')
with torch.no_grad():
logits = model(**tokens).logits
probs = softmax(logits)
label = LABELS[argmax(probs)]
confidence = max(probs)
IF GEMINI_API_KEY:
mh_label = call_gemini(request.text)
```

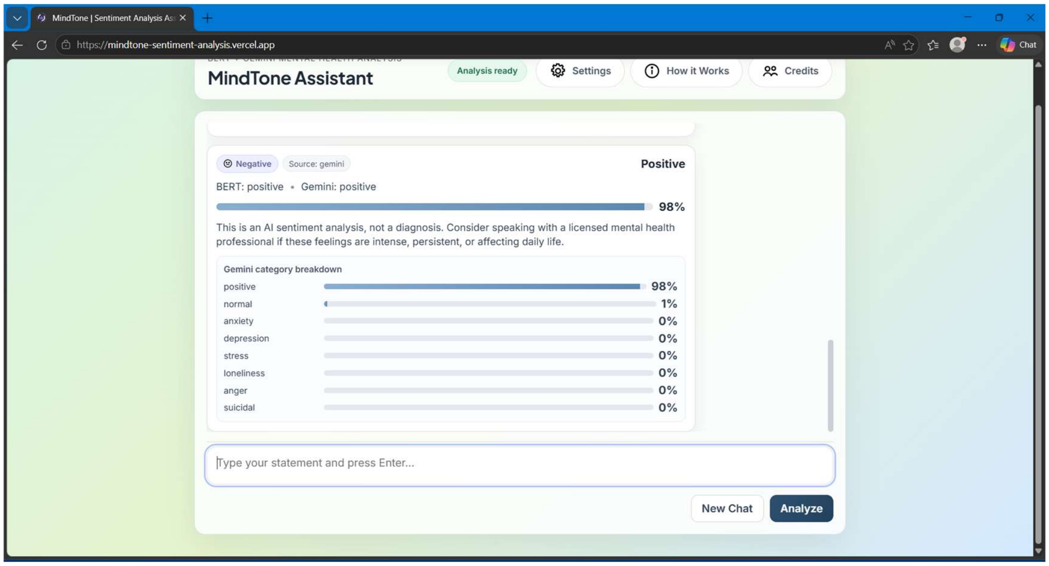
STORE (text, label, confidence, mh_label) in SQLite

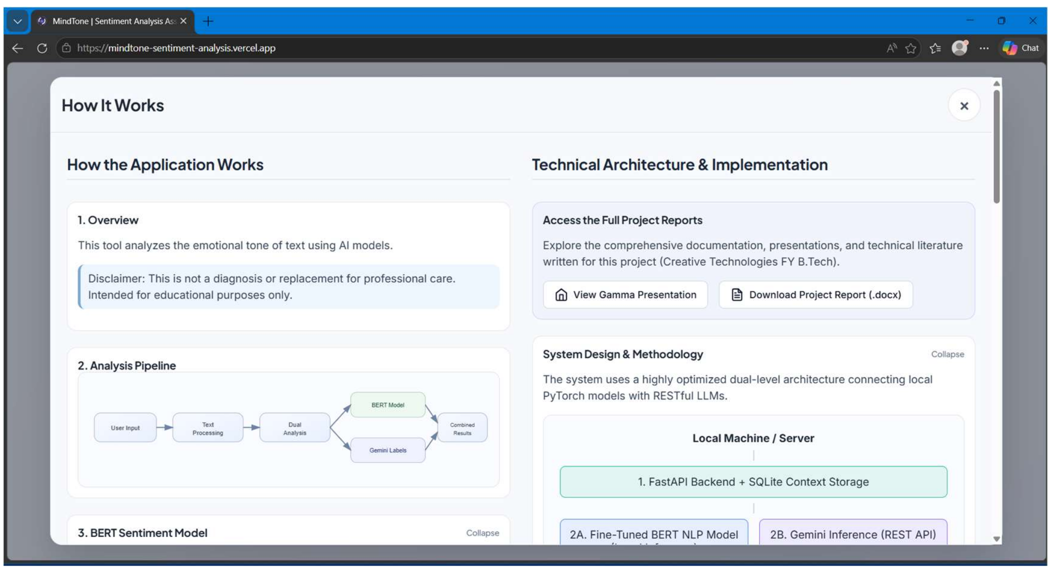
RETURN {sentiment, confidence, mental_health_label}

APPENDIX A2 - PROJECT PHOTOGRAPHS






MindTone | Sentiment Analysis As
https://mindtone-sentiment-analysis.vercel.app
MindTone Assistant
Analysis ready
Settings
How it Works
Credits
Negative
Source: gemini
Positive
BERT: positive • Gemini: positive
98%
This is an AI sentiment analysis, not a diagnosis. Consider speaking with a licensed mental health professional if these feelings are intense, persistent, or affecting daily life.
Gemini category breakdown
positive 98%
normal 1%
anxiety 0%
depression 0%
stress 0%
loneliness 0%
anger 0%
suicidal 0%
Type your statement and press Enter...
New Chat
Analyze


MindTone | Sentiment Analysis As
https://mindtone-sentiment-analysis.vercel.app
How It Works
How the Application Works
Technical Architecture & Implementation
1. Overview
This tool analyzes the emotional tone of text using AI models.
Disclaimer: This is not a diagnosis or replacement for professional care. Intended for educational purposes only.
Access the Full Project Reports
Explore the comprehensive documentation, presentations, and technical literature written for this project (Creative Technologies FY B.Tech).
View Gamma Presentation
Download Project Report (.docx)
2. Analysis Pipeline
User Input
Text Processing
Dual Analysis
BERT Model
Gemini Labels
Combined Results
System Design & Methodology
Collapse
The system uses a highly optimized dual-level architecture connecting local PyTorch models with RESTful LLMs.
Local Machine / Server
1. FastAPI Backend + SQLite Context Storage
2A. Fine-Tuned BERT NLP Model
2B. Gemini Inference (REST API)
3. BERT Sentiment Model
Collapse

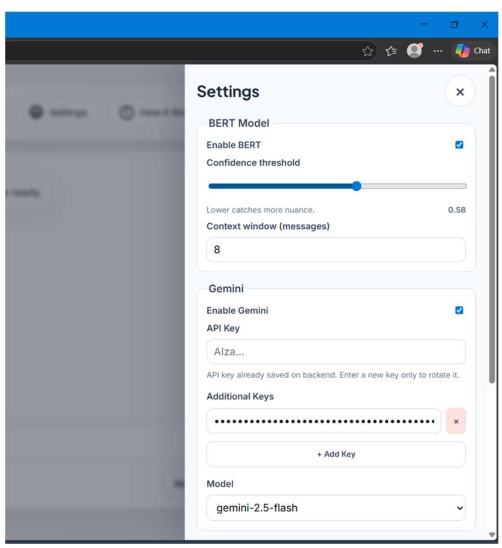
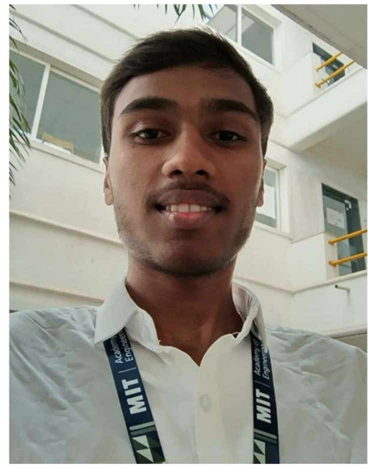

Chat
Settings
BERT Model
Enable BERT
Confidence threshold
Lower catches more nuance.
0.58
Context window (messages)
8
Gemini
Enable Gemini
API Key
AIza...
API key already saved on backend. Enter a new key only to rotate it.
Additional Keys
+ Add Key
Model
gemini-2.5-flash

Table of Students / Name / PRN:

Sr. No.	Name	PRN	Photograph
1	Ajinkya Magar	202501100063	
2	Aarsh Joshi	202501040435	
3	Varad Kurwade	202501100019	

APPENDIX A3 - PROJECT LINKS

Link for Project Video Uploaded:

https://drive.google.com/drive/folders/1k1befv5XD5O2FCbw7Dd_kQPsb_nhIHze?usp=sharing

Working Project Link (After Deployment):

https://mindtone-sentiment-analysis.vercel.app/

Link of GitHub Repository:

https://github.com/ajinkyamagarphys-jpg/sentiment-analysis-model.git